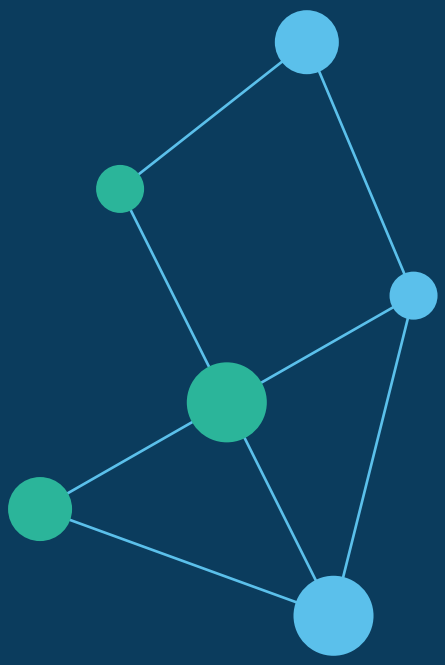

Dataset Details

AI-Powered Telecom Network Operations Assistant

Data used for retrieval, baselines and prediction

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1. Dataset Overview

The project uses a tabular 5G network operations dataset. It supplies the historical incidents that are searched and ranked, the records used for historical averages, and the data used to train and test the prediction models.

Item	Detail
Public source	Hugging Face dataset fenar/5g_netops (https://huggingface.co/datasets/fenar/5g_netops)
Project file	data/telecom_network_incidents_with_id.csv
Rows	100,000 (single train split)
Missing values	None (0%), according to schema.jsonl and incident_knowledge.jsonl
Columns	27 fields (listed in Section 2)
Data type	Tabular, CSV format
License (as shown on the dataset page)	apache-2.0
Nature of data	Synthetic. The project documents describe the data as synthetic.

The Hugging Face page lists 27 fields, which matches the column count in the project schema. The project file name includes "with_id", and evaluation output refers to incident IDs such as INC058190. The public page lists no ID column, so the ID appears to be added in the project copy. This is an assumption. Confirm it in the schema files.

2. Field Dictionary

Types, units and ranges come from schema.jsonl and agree with the column summaries in the Hugging Face dataset viewer. Ranges are the minimum and maximum values. The groups are an organisation used in this document.

2.1 Context and geography

Field	Type	Values / range
Season	string	4 values: Winter, Fall, Spring, Summer
City	string	38,066 distinct values
State	string	59 distinct values (state and territory codes)
Zip	integer	503 to 100,000
Issue Reported	string	2 values: yes, no

2.2 Performance

Field	Type	Range
Cell Availability (%)	float	74.3 to 99
Throughput (Mbps)	float	115 to 1,280
Latency (ms)	float	4.62 to 176
Packet Loss Rate (%)	float	0.1 to 17
Call Drop Rate (%)	float	0 to 6.9
Handover Success Rate (%)	float	90 to 100
Jitter (ms)	float	0.1 to 50
Connection Setup Success Rate (%)	float	90 to 99

2.3 Operations

Field	Type	Range
MTTR (hours)	float	0.5 to 10
Alarm Count	integer	0 to 26
Critical Alarm Count	float	0 to 13.9
Parameter Changes	integer	0 to 19
Successful Configuration Changes (%)	float	90 to 99

2.4 Load

Field	Type	Range
Data Usage (GB)	float	10 to 50
User Count	integer	100 to 10,000

2.5 Signal and environment

Field	Type	Values / range
Signal Strength (dBm)	float	-120 to -60
Temperature (°C)	float	-10 to 40
Humidity (%)	float	20 to 90
Weather	string	4 values (sample rows show Clouds, Clear, Rain, Snow)

2.6 Security

Field	Type	Range
Security Incidents	integer	1 to 19
Authentication Failures	integer	0 to 63

2.7 Prediction target

Field	Type	Range
Fault Occurrence Rate (%)	float	0 to 81.72 (mean 38.16, median 39.46, std 14.03)

Throughput (Mbps), listed under Performance, is the target of the network performance model. In schema.jsonl, Fault Occurrence Rate (%) is marked as the fault risk prediction target (regression). Statistics are from incident_knowledge.jsonl.

2.8 Missing values

schema.jsonl and incident_knowledge.jsonl both report no missing values (0 missing, 0.0%).

3. Sample Record

First row shown in the dataset viewer.

Field	Value	Field	Value
Season	Winter	Alarm Count	13
Cell Availability (%)	98.045	Critical Alarm Count	6.71802
MTTR (hours)	3	Parameter Changes	18
Throughput (Mbps)	653.5314	Successful Config. Changes (%)	91.85
Latency (ms)	74.336	Data Usage (GB)	47.49
Packet Loss Rate (%)	6.4375	User Count	2,442
Call Drop Rate (%)	1.466671	Signal Strength (dBm)	-110.12
Handover Success Rate (%)	99.29	Jitter (ms)	15.67
Connection Setup Success (%)	99	Security Incidents	8
Authentication Failures	2	Temperature (°C)	-6.75
Humidity (%)	28.79	Weather	Clouds
Issue Reported	yes	City / State / Zip	Schneiderville / MN / 46,530
Fault Occurrence Rate (%)	33.582417		

4. How the Dataset Is Used

Use	How the data is used
Metadata filtering	Query plan filters by city, state, season, weather, incident ID and ZIP code. Numeric conditions are applied only when the user states them.
Vector retrieval (ChromaDB)	Incident text is embedded and stored in a persistent local Chroma collection for semantic search.
Lexical retrieval (BM25)	The same incident text is indexed with BM25 for keyword search. Semantic (about 60%) and BM25 (about 40%) scores are combined with metric relevance scoring.
RCA	The RCA agent uses the top-ranked retrieved incidents as its evidence. Output is a hypothesis.
Historical averages	Descriptive averages are calculated over metadata-filtered records when a query asks for averages, performance, trends, assessment or impact.
Classification	Incident priority (P1 to P4) uses dataset-relative prototype thresholds. These are not telecom SLAs or 3GPP limits.
Machine learning	The throughput and fault-occurrence models are trained on 80,000 rows and tested on 20,000 held-out rows.

5. Context Files in Synapt Context Substrate

Four JSONL files are loaded into the Synapt Context Substrate layer. Synapt serves them to the assistant through MCP: domain context for the Query agent and SOP procedures for the Recommendation agent.

File	What it holds	Source
`domain_knowledge.jsonl`	Telecom concepts with definitions, units and related metrics, plus simple relationships between them	Hugging Face dataset
`incident_knowledge.jsonl`	How historical incident records are represented and retrieved, with field groups, allowed values and rules	Hugging Face dataset
`schema.jsonl`	The 27 columns with type, unit, description and range, plus the ML target	Hugging Face dataset
`sop_knowledge.jsonl`	11 troubleshooting procedures (SOP_001 to SOP_011)	**Synthetically created**

The SOP knowledge file is synthetic. It is not vendor-specific and must not be treated as an operator's approved procedure. The other three files come from the Hugging Face dataset.

5.1 domain_knowledge.jsonl

- **Concepts (7):** latency, packet loss, call drop, handover, network congestion, signal strength and network fault.
- **Relationships (5):** congestion can contribute to high latency and to packet loss; poor signal strength and poor handover can contribute to call drops; critical alarms indicate a potential network fault.
- The wording is "can contribute to". These are possible links, not proven causes.

5.2 incident_knowledge.jsonl

- Describes the dataset as 100,000 rows and 27 columns with no missing values.
- Groups fields into: identity and context, network performance, operations and repair, usage and load, signal and environment, security, and the fault measure.
- Lists 25 retrieval fields, the allowed values (4 seasons, 4 weather types, issue reported yes/no, 59 states, 38,066 cities) and Fault Occurrence Rate statistics.
- Retrieval method: semantic similarity over the original dataset, which is the source of truth.

Rules stated in the file:

- Do not infer a definitive root cause from a historical observation unless the dataset has a root-cause field.
- Do not invent corrective actions or troubleshooting steps from this file.
- Do not treat correlation between metrics as proven causation.
- Use the original dataset values as the source of truth.

5.3 schema.jsonl

- Dataset name telecom_network_incidents, 100,000 rows, 27 columns, no missing values.
- Each column has a type and description. Numeric columns also have a unit and a minimum and maximum.
- ML target: Fault Occurrence Rate (%), a regression task (fault risk prediction).

5.4 sop_knowledge.jsonl (synthetic)

Each SOP has a scenario, trigger metrics, diagnostic checks and recommended actions.

SOP ID	Name	Scenario
SOP_001	High Latency Investigation	High network latency
SOP_002	Packet Loss Investigation	Elevated packet loss
SOP_003	Call Drop Investigation	Elevated call drop rate
SOP_004	Poor Handover Performance	Low handover success rate
SOP_005	Low Signal Strength Investigation	Poor cellular signal strength
SOP_006	Network Congestion Investigation	Possible network load or congestion
SOP_007	Alarm and Fault Investigation	High alarm or critical-alarm activity
SOP_008	Configuration Change Investigation	Degradation linked to configuration activity
SOP_009	Network Availability and Recovery Investigation	Low cell availability or long repair time
SOP_010	Security and Authentication Investigation	Elevated security incidents or authentication failures
SOP_011	Multi-Metric Network Degradation Investigation	Several indicators degraded at once

Usage rules stated in the file:

- Retrieve SOPs based on the observed condition and supporting metrics.
- Use SQL, ML and RAG evidence before choosing a recommendation.
- Do not claim an action will definitely resolve an issue.

- If evidence is insufficient, recommend investigation and escalation instead of inventing a root cause.
- Adapt recommendations to the affected metrics and the retrieved historical incidents.

Synapt domain context and SOP procedures are advisory. Historical incidents from the dataset remain the only evidence source.

6. Embedding Details

Item	Setting
Library	sentence-transformers
Model	BAAI/bge-small-en-v1.5, loaded locally
Vector store	ChromaDB (persistent local collection)
Lexical index	rank-bm25 (BM25)
Schema mapping	config/metric_schema.py maps column headers and aliases to stable snake_case field names

The exact text template used to turn a row into the incident text is defined in the retrieval code and is not described here.

7. Fields Used for Prediction

The prediction models use performance, operational, load, signal, environmental, security and geography features, plus season and weather. The exact required feature list is set in config/prediction_config.py, which was not part of the material used for this document, so individual required fields are not listed here.

Model	Target	Notes
Network performance model	Throughput (Mbps)	Runs only when the full required feature set is supplied.
Fault tolerance model	Fault Occurrence Rate (%)	Mapped to a risk tier: LOW, MEDIUM, HIGH, CRITICAL. The performance model's output is used as an additional input.

Telemetry must be complete, numeric and finite. Percentages must be from 0 to 100. Counts, users, data usage and duration must be non-negative. If a check fails, the prediction is rejected or marked incomplete.

8. Why Historical Averages Are Baselines

A historical average describes what the stored records show for a filtered group, for example one city in one season. It is not a measurement of the network right now, and it is not a forecast. A prediction is a model estimate from current telemetry that the user supplies. For this reason the interface labels averages as a **historical baseline** and shows a prediction only when valid current telemetry is provided. A baseline is never presented as a prediction.

9. Data Quality and Limitations

Topic	Note
Synthetic data	Records are synthetic. Patterns, retrieval quality and model scores may not carry over to real operator data.
Narrow ranges	Several fields have bounded ranges (for example handover success 90 to 100%). Baselines and models reflect this dataset only.
No timestamps	The dataset has no time-series timestamps, so it cannot show trends over time. Scenario comparison is a what-if view, not a time trend.
Missing values	None reported (0%) in the schema and incident knowledge files.
Privacy	The fields contain location data (city, state, ZIP) but no subscriber-level identifiers. City names in the sample look fictional.
Bias	The spread of records across states, seasons and weather types has not been analysed in the material used here, so coverage gaps are unknown.
Production use	The system is not connected to live telecom telemetry, and the SOP examples in the project are synthetic or demonstration data.

10. Source Files and References

- Hugging Face dataset page: fenar/5g_netops (field names, types, ranges, row count, license, sample row).
- Technical Stack document (dataset size, embedding model, retrieval and ML libraries).
- Project architecture brief (retrieval design, classification, validation rules).
- Context files loaded into Synapt: domain_knowledge.jsonl, incident_knowledge.jsonl, schema.jsonl, sop_knowledge.jsonl.
- Not available: config/metric_schema.py and config/prediction_config.py.